CAT/XSD Group Name: LS-CAT (Life Sciences Collaborative Access Team)

CAT Director/Group Leader:

Scientific Director: Prof. Alfonso Mondragón
Manager: Keith Brister

Provide a numerical rating (1= exceptional performance, 2 = expected level of performance; 3 = weak performance, improvement needed) and brief comments for each of the following:

1. Scientific Mission and Impact – 1.5
2. Beamline Productivity – 1.5
3. Leveraging of specific APS characteristics – 1.5
4. Effectiveness of beam time use – 2
5. General User program quality – 1.5
6. Adequacy and reliability of funding – 1
7. Future Plan – 1.5
8. Overall – 1.5
9. Recommended Actions including recommendation to APS for continuation, a re-review before five years, or termination/decommissioning. (no numerical score given)

Specific comments are given in the sections below.

As presented in the detailed report below, the committee strongly recommends continuation of the LS-CAT sector.

Report of LS-CAT Review, November 4, 2014

Submitted January 22, 2015

Review Panel

Paul D. Adams (Lawrence Berkeley National Laboratory, Physical Biosciences Division)
Britt Hedman (Stanford University, Stanford Synchrotron Radiation Lightsource, SAC, Chair)
Christopher P. Hill (University of Utah School of Medicine, Department of Biochemistry)
Lisa Miller (Brookhaven National Laboratory, National Synchrotron Light Source-II)

Review Agenda

The Scientific Advisory Committee (SAC) review of the insertion device beam lines of Sector 21, operated by LS-CAT, took place on November 4, 2014 by the panel members listed above. After an overview of life sciences at APS by Robert Fischetti (jointly for the two biosciences reviews that took place the same day), the specific LS-CAT review program took place at Sector 21. It was attended by the LS-CAT sector management and staff, several invited scientists, Chris Jacobsen, APS Associate Division Director, and Susan White-DePace, APS User Administration. The following topics were covered in the agenda:

- Overview of LS-CAT's science, history and organization – Alfonso Mondragón
- LS-CAT design and user program – Keith Brister
- Two science talks by CAT member representatives: "Biological methane oxidation" by Amy Rosenzweig, Northwestern University, and "Structural mechanisms of DNA base excision and replication repair" by Brandt Eichman, Vanderbilt University
- LS-CAT enabling technologies – Keith Brister
- The future of LS-CAT – Keith Brister and Alfonso Mondragón
- Bionanoprobe science – Gayle Woloschak
- A discussion between reviewers and sector staff (without the LS-CAT management) was held during a working lunch.
- The agenda was closed with an executive session, followed by a close-out with Alfonso Mondragón and Keith Brister.

Introduction and Background

The Life Sciences Collaborative Access Team (LS-CAT) was formalized in 2002, with seed funding from the State of Michigan, and with Northwestern University as managing partner. The first beam to the sector beam lines was delivered in December 2006. The sector beam lines use two canted undulators: 21-ID-D provides tunable beam (also has white beam capability) with 10-50 µm beam size for protein crystallography. The 21-ID-D beam line is shared (25%) with a bionanoprobe (added 2012) that uses 50 nm beams for hard x-ray fluorescence microscopy. The nanoprobe 25% beam time constitutes the general user time for the sector. 21-ID-F and 21-ID-G are Se SAD non-tunable protein crystallography beam lines using the second undulator as source. They are in operation simultaneously through the use of Laue diamond monochromators. A fourth line, 21-ID-E is used as development line.

LS-CAT is organized as a consortium, currently with 7 university/institute member institutions, having half or full "shares", with an 8th potentially (re)joining. There are 7 partners recognized as Associate Members, ranging from universities to pharmaceutical industry. The Associate Members partner with one of the regular members.

The consortium management structure is based on a Management Board that represents the LS-CAT institutions, and that reviews the performance, sets policies and approves the budget. The LS-CAT Science Director reports to the board. All LS-CAT staff members are Northwestern University employees.

The LS-CAT science program is dedicated to structural studies of a wide range of problems in biochemistry and biomedical sciences, ranging from fundamental understanding to drug discovery. There is also use for structural genomics projects (NIGMS PSI, which will terminate in summer 2015, and NIAID, currently active through 2017). There is a significant amount of industrial use.

Scientific Productivity

A distinctive feature of LS-CAT is its commitment to providing excellent service to users for the collection of macromolecular x-ray diffraction data. Success is reflected in the evidently very happy and loyal users, which was manifest from the presentations at the review session and the willingness of partner institutions to maintain their financial commitment. The high productivity and user satisfaction is supported by the strong commitment to training, including intensive workshops. It also reflects a flexible scheduling system that is effective in assigning beam time with short turn-around time.

High productivity is indicated by the metrics for publications, PDB depositions, and contributions to Ph.D. theses. LS-CAT is one of the most prolific APS sectors for publications, second only to NE-CAT, with, for example, 142 publications in 2012. The impact of these papers, as indicated by this panel (very imperfectly) by the fraction of papers in journals of the highest impact factor, is seemingly a bit lower than most other leading synchrotron facilities, but nevertheless includes 25 papers in Cell, Nature, or Science. Presentations given at the review reinforced the impression that work being performed at LS-CAT is of central importance to very strong research programs.

The number of PDB depositions is very strong at a total of 1190 from 2011-2014, and seems to be the highest of APS sectors and for a number of other productive ID ports internationally. This is especially notable given the relatively high use of LS-CAT for data collection by industrial partners, who tend not to publish or deposit coordinates.

A total of 108 Ph.D. theses have credited LS-CAT since 2007. This is a notably high number that dwarfs the number theses reported for other APS sectors. This is consistent with the evident commitment to education, training and service, and is a notable success.

Although productivity is already high, the LS-CAT leadership recognizes that there may be opportunities to further improve, especially as technology continues to develop. For example, in principle, nighttime scheduling could be made more efficient. While this is a potential opportunity that might be exploited, it may require a collaborative approach to beam-line support that is not encouraged by the APS management structure, and it is not currently clear that there is a substantial reserve of unmet user demand. On the other hand, the staff members perform change-overs and user support on weekends.

The review panel concludes that the scientific productivity is excellent for the macromolecular crystallography program at the sector and, given additional funding, this could even expand further, for example by providing the ability to run 24/7.

Selected metric details include: The 21-ID-D station produces 2000-2300 data sets per year, resulting in 200-370 structures and around 50 publications, of which about 20% are in high-impact journals.

The 21-ID-F stations produces 2900-3200 data sets, 115-130 structures, and 50-90 papers per year, while the 21-ID-G station provides about 2500-2800 data sets, 90-135 structures, and 40-60 publications per year. Industrial and structure genomics users contribute to the large number of data sets.

The number of unique users is growing with FY14 data showing 351, 210, and 136 unique users for 21-ID-D/-F/-G, respectively.

Bionanoprobe: With its recent implementation and commissioning, there is not yet an established productivity, but at the review it was noted that a number of publications are anticipated, and based on the presented results, the scientific impact will be significant. This instrument is used twice per run cycle, and while unique and at the very forefront for this technique and science, takes away beam time (and thus efficiency/productivity) for the highest performing 21-ID-D crystallography beamline. As described in the "Future" section, this might be addressed at the time of the APS upgrade by moving this instrument to 21-ID-E.

Management Structure

Northwestern University is the legal entity for the LS-CAT and the managing partner of the 7-member CAT.

The management of the LS-CAT is through a Management Board that has a representative from each of the LS-CAT member institutions. The Board reviews the LS-CAT beam line performance and the LS-CAT policies semi-annually and approves the budget annually. There is an LS-CAT Scientific Director (Alfonso Mondragón) and an LS-CAT Manager (Keith Brister, PI for the funding), and both report to the Board. This structure ensures that the needs of the member institutions are being incorporated and met.

LS-CAT reports within the Northwestern University administration, and accordingly, all LS-CAT staff members are Northwestern University employees.

The operations are overseen by the Northwestern University Synchrotron Research Center (which also includes DND-CAT) that, in turn, reports to the Northwestern University Vice President for Research.

In addition to Mondragón and Brister as managers, the staffing consists of five scientific staff covering collaborative research, user support, instrumentation, computing, network, training and dissemination, among other things. They are supported by two technical staff members and an administrator (scheduling, finance, administration). The staff members work as an integrated team, distributed about evenly across the three beam lines, and with significant cross-training. At the review, the staff showed great enthusiasm and responsibility for scientific user support, keeping the instrumentation at high performance, and training. They carry a substantial workload and professed a high level of satisfaction with the organization and management. The staff and management are to be congratulated for providing such a smoothly and efficiently running operation, with very satisfied users and high scientific productivity.

The bionanoprobe is supported by two part-time APS staff – an estimate of ~50% FTE total was provided as level of effort. Each time the instrument is moved in/out of the station, the setup needs to be optimized. This is scheduled twice per run cycle, with the process/scheduling coordinated with non-LS-CAT APS management, as this is part of the APS General User program.

Current Capabilities

The LS-CAT sector consists of four beam lines, served by two canted undulators. Of these beam lines, three (21-ID-D, 21-ID-F, and 21-ID-G) are in active use. The remaining beamline (21-ID-E) is currently used for development, and could be put into active service at relatively modest cost. The LS-CAT team is to be congratulated for developing an ID-only sector where the beam lines are heavily used, in contrast to other sectors where bend magnet beam lines have typically received little use. The LS-CAT sector provides a fully tunable source (21-ID-D), which has white beam capability. The other two beam lines (21-ID-F and 21-ID-G) provide monochromatic sources of x-rays from Laue diamond monochromators, tuned to enable Se-SAD capability.

The end stations provide robotic sample mounting hardware, high performance goniometers and x-ray detectors. These facilities are competitive with other high performance undulator beam lines at the APS and other 3rd generation sources. As a result the beam lines have been very productive, enabling high quality research and a large number of PDB depositions. Additionally, the beam lines have been designed and operated in a manner that supports high quality service to users with high reliability.

However, the LS-CAT leadership has recognized that beam line upgrades are necessary to continue to best support user research. A new high performance Eiger x-ray detector has been funded by the CAT partners, purchased from Dectris, and will be deployed on 21-ID-D when delivered. This will be either a 9M or 16M device depending on the requirements closer to delivery. The review committee suggests focusing on a 9M device in order to best support user science while also retaining funds to support other beam line upgrades. As a result of the Eiger purchase, the LS-CAT management recognizes that the computing and networking infrastructure will need to be upgraded to support the accelerated rate of data collection. This should be a high priority. The next priority is an upgrade of the robotic crystal automounter systems on each beam line. Currently CATS robots are used in combination with Spine pins and vials. Although operating at 99% reliability, this is problematic for user operation and beam line staff support. Management has addressed this with planning underway. A move to an ALS/Unipuck Dewar and an upgraded pin gripper will do much to address this situation. The potential impact of this upgrade is high, and is likely to reduce staff support effort and enable routine remote data collection during the night shifts. This upgrade should be the second highest priority beyond the network/computing upgrades.

Adequacy and Reliability of Funding

As mentioned, the LS-CAT is funded by 7, and possibly soon to be 8, partners. These partners have committed to “shares” that provide the operations funding for the beamlines. They also provide one-time investment funding, such as for the new Eiger detector.

Adequacy: The current operations funding is adequate, but the CAT would benefit from increased funding that would allow for additional and/or upgraded hardware, such as robotics, and additional support staff to enable 24/7 operations.

Reliability: Funding is in hand through 2017. As described below, the continuation of funding for LS-CAT depends on whether the partners will renew/extend the agreements for a new defined multi-year term. There might also be an impact by the APS upgrade shutdown, as is also further described below. The reliability of the funding is in many ways higher than for federally funded CATs, and the consortium has been stable for a very long time, which provides high confidence for reliable future funding.

Future Directions

There are two primary uncertainties in the future of the LS-CAT program. First, the current 15-year agreement between the members expires in February 2017. The management has already begun discussions, and it appears likely that the member agreement will be extended for another five years. Second, the plans for the APS upgrade and its timeline are in flux so it is currently unclear how it will impact the LS-CAT program. In the short term (< 5 years), upgrades to keep the beamlines state-of-the-art are planned, including larger diameter Be lenses on the F and G stations, a new Eiger detector for the D station, and an upgrade to the computing infrastructure. An additional upgrade to more reliable robotics would greatly improve beamtime efficiency because users cannot currently use the robot overnight due to potential failures. LS-CAT staff are currently evaluating ways to improve the robotics reliability, e.g. by adopting an ALS-style Dewar design. This should be a high priority. Another possible upgrade is the implementation of a multilayer monochromator of the D station, which would improve the flux for both the bionanoprobe and the MX programs.

The longer term (>5 years) plans for LS-CAT will very much depend upon the timeline and parameters for the APS upgrade. With the APS upgrade (as planned today), LS-CAT could see an increase in brightness by a factor of 100 if the insertion devices for the beamlines are also upgraded. This would greatly improve the MX programs, but perhaps the largest effect would be in the performance of the bionanoprobe. In addition, the upgrade would allow LS-CAT to put the bionanoprobe on a dedicated beamline (i.e. 21-ID-E) rather than sharing with the MX program on 21-ID-D. Based on the oversubscription of other lower resolution micro/nanoprobes at APS, it is clear that the bionanoprobe program will grow dramatically over the next few years and it will need more than 25% of the beamtime (the current allocation) to be productive.

In order to carry its members during the APS dark period, LS-CAT is already considering how to provide beamtime and support to their users at other synchrotron facilities. Because of this expected dark period, and the upcoming renewal of the member agreement, it is crucial for APS management to keep the LS-CAT team up to date with the upgrade plans and timeline in order to provide for a smooth transition for the LS-CAT users.

Conclusion

The review panel concludes that the LS-CAT is an exceptionally highly productive CAT that provides excellent user support at well-performing beamlines. The users are very satisfied and productive, and the staff members display a positive can-do attitude. The LS-CAT members represent very strong scientific programs. LS-CAT has a dedicated educational/training component with a high number of Ph.D. theses as well as training for undergraduates, and intense user training workshops. The CAT focuses mainly on operations. R&D is sometimes challenging due to limitations in funding – but is well executed through collaborations as appropriate.

LS-CAT is well managed, and has a strong scientific and technical leadership. Funding is currently stable and adequate.

The nano-probe is a unique instrument in the U.S. at present. It currently uses 25% time on 21-ID-D and will likely merit additional beam time in the future. This anticipated increase in demand should to be strategically addressed at the time of the APS upgrade.

The review committee strongly recommends a continuation of the LS-CAT at sector 21-ID.